INCH-POUND
MIL-DTL-27072/83C
w/AMENDMENT 1
20 May 2016
SUPERSEDING
MIL-DTL-27072/83C
12 March 2012

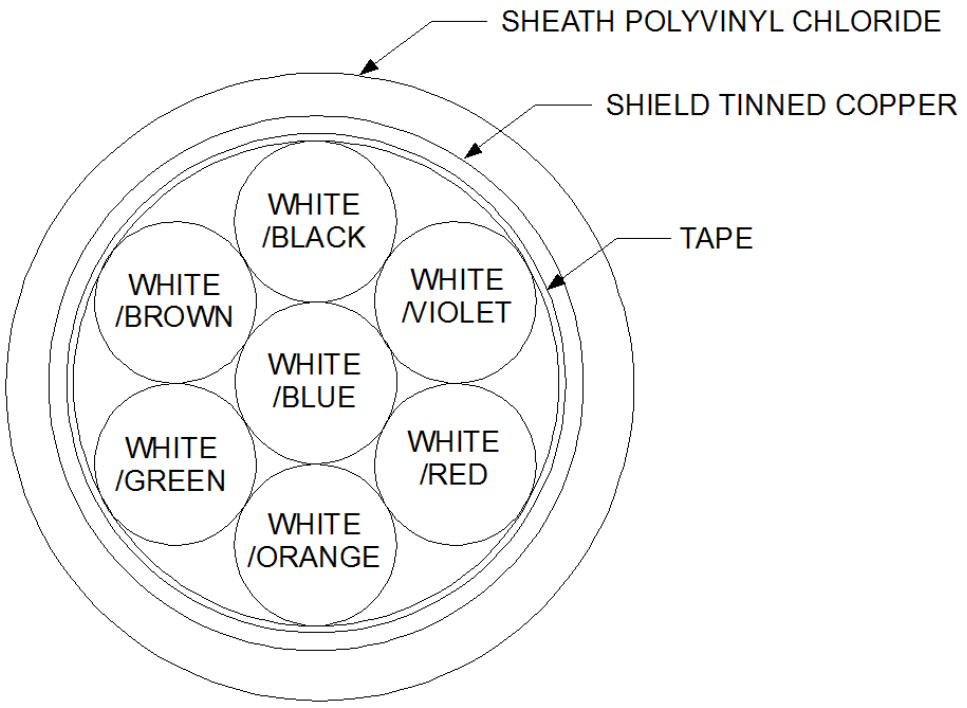
DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR (7 CONDUCTORS) AND SINGLE SHIELDED, M27072/83MACA*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27072.

REQUIREMENTS:



NOTE: The complete Part or Identifying Number (PIN) requires a color code designator (see table I).

FIGURE 1. Cable configuration.

Design and construction:	See figure 1.
Number of conductors:	7
Temperature rating:	105 °C

AMSC N/A

FSC 6145

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Components:
- Basic wire: NEMA HP 7-BN-BFB* (see table I)
  - Conductor: Tinned copper
    - Primary insulation: Polyvinyl chloride
      - Colors: White with blue tracer; White with brown tracer; White with green tracer; White with orange tracer; White with red tracer; White with violet tracer; White with black
    - Jacket over insulation: Polyamide
      - Color: Natural

Filler: Not applicable

Cabling: Not applicable

Binder tape material: In accordance with MIL-DTL-27072

Shield:
- Material: Tinned copper
- Gauge: 36 AWG
- Braid angle: Braid 20 to 40 degrees

Jacket undershield: Not applicable

Sheath:
- Material: Polyvinyl chloride
  - Wall thickness: .025 inch nominal
  - Tensile strength: In accordance with MIL-DTL-27072
  - Ultimate elongation: In accordance with MIL-DTL-27072
  - Accelerated aging: In accordance with MIL-DTL-27072
  - Color: See MIL-DTL-27072

Finished cable: See table I
- Cold bend: Mandrel size in accordance with MIL-DTL-27072
- Dielectric withstanding voltage: 2,000 V rms
- Conductor resistance: In accordance with basic wire specification

TABLE I. Construction data.

PIN 27072/83	Basic wire except color NEMA HP 7-BN-BFB*1/,2/ (reference)				Finished cable diameter inch(mm) ±10%
	Wire size (AWG)	Number of strands	Strand size	Nominal diameter inch(mm)	
MACA*	22	7	30	.056(1.42)	.247(6.27)

1/ Asterisks (*) shall be replaced by the appropriate color designator.
2/ Dimensions are in inches. Metric equivalents are given for information only.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced document. This document references MIL-DTL-27072.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 6145-2016-028)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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